AWS VPC Peering

Final Project Report

1. Project Details

Field	Details
Project title	AWS VPC Peering
Student name	Sabitha Ramu
Batch	1
AWS Region	ap-southeast-2 (Asia Pacific – Sydney)
Completion date	23-10-2026
Course	EWB Courses – AWS VPC Peering Student Project

2. Architecture Diagram

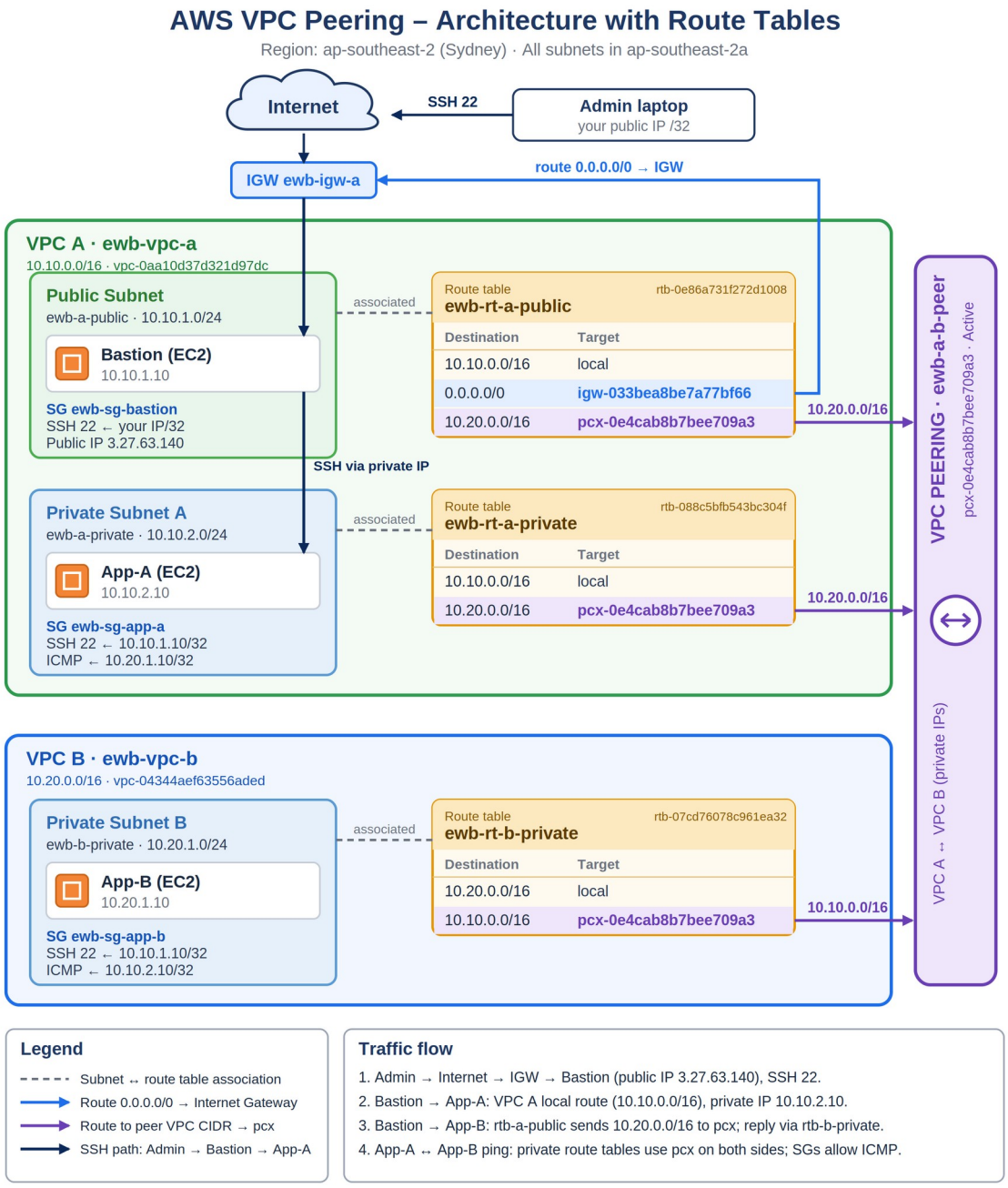


Figure 1: AWS VPC Peering architecture

Purpose of the VPCs: VPC A (ewb-vpc-a, 10.10.0.0/16) hosts the Bastion in a public subnet and App-A in a private subnet. VPC B (ewb-vpc-b, 10.20.1.0/24 private subnet inside 10.20.0.0/16) hosts App-B, which is a private-only workload.

Private IP connectivity: The two VPCs are joined by the ewb-a-b-peer peering connection, so App-A and App-B talk to each other using private IPs only, without traffic going over the internet.

Bastion usage: The Bastion is the only host with a public IPv4 address. Admin access is via SSH to the Bastion public IP, and from the Bastion the private IPs of App-A and App-B are used for onward SSH.

3. Resource Inventory

Region: ap-southeast-2 (Sydney). Names and CIDRs are as per the project design; IDs are from the AWS Console.

Resource	Name / CIDR	AWS Resource ID
VPC A	ewb-vpc-a — 10.10.0.0/16	vpc-0aa10d37d321d97dc
VPC B	ewb-vpc-b — 10.20.0.0/16	vpc-04344aef63556aded
Public subnet A	ewb-a-public — 10.10.1.0/24 (ap-southeast-2a)	subnet-0c7d52346183128ca
Private subnet A	ewb-a-private — 10.10.2.0/24 (ap-southeast-2a)	subnet-0f5ddec5de6162579
Private subnet B	ewb-b-private — 10.20.1.0/24 (ap-southeast-2a)	subnet-0306c77e066797ff0
Internet Gateway (VPC A)	ewb-igw-a	igw-033bea8be7a77bf66
Bastion	10.10.1.10	i-0b21ded55c6de57ec Public IP: 3.27.63.140
App-A	10.10.2.10	i-08c88dc418e5bc4cb
App-B	10.20.1.10	i-063ea200450a79cee
Peering	ewb-a-b-peer — Active	pcx-0e4cab8b7bee709a3
Route table (public A)	ewb-rt-a-public	rtb-0e86a731f272d1008
Route table (private A)	ewb-rt-a-private	rtb-088c5bfb543bc304f
Route table (private B)	ewb-rt-b-private	rtb-07cd76078c961ea32
Security group – Bastion	ewb-sg-bastion	sg-0a5d09a13a83664df
Security group – App-A	ewb-sg-app-a	sg-0e167c589b25df2ef
Security group – App-B	ewb-sg-app-b	sg-02bc4845739270db9

4. Route Tables and Subnet Associations

Each route table is shown below with its configured routes and the subnet it is associated with.

4.1 ewb-rt-a-public (rtb-0e86a731f272d1008)

Associated subnet: Public subnet A (10.10.1.0/24)

Destination	Target
10.10.0.0/16	local
0.0.0.0/0	Internet Gateway (igw-033bea8be7a77bf66)
10.20.0.0/16	Peering (pcx-0e4cab8b7bee709a3)

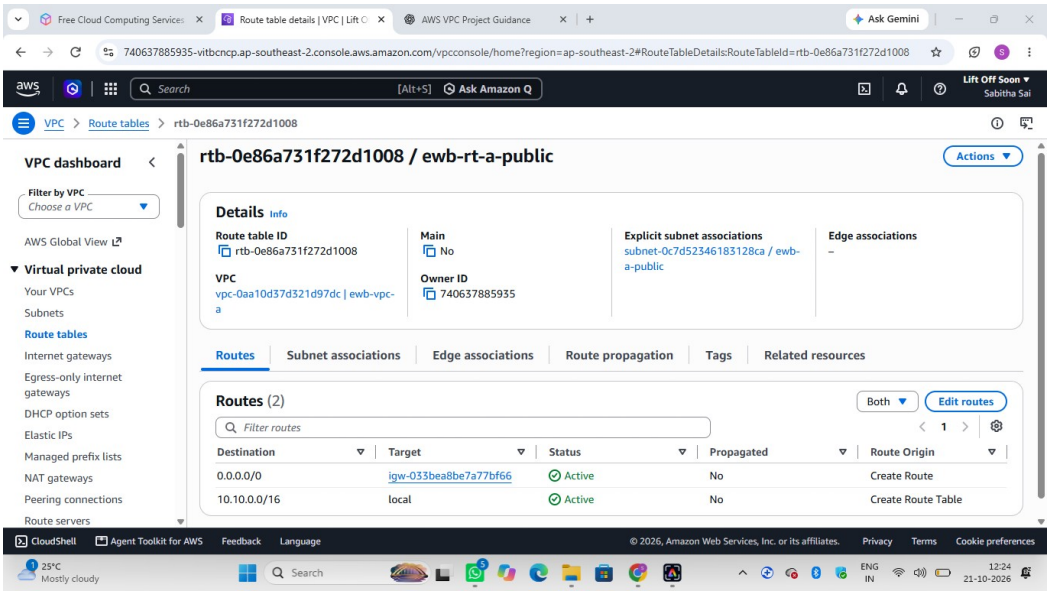


Figure 4.1a: ewb-rt-a-public – Routes tab

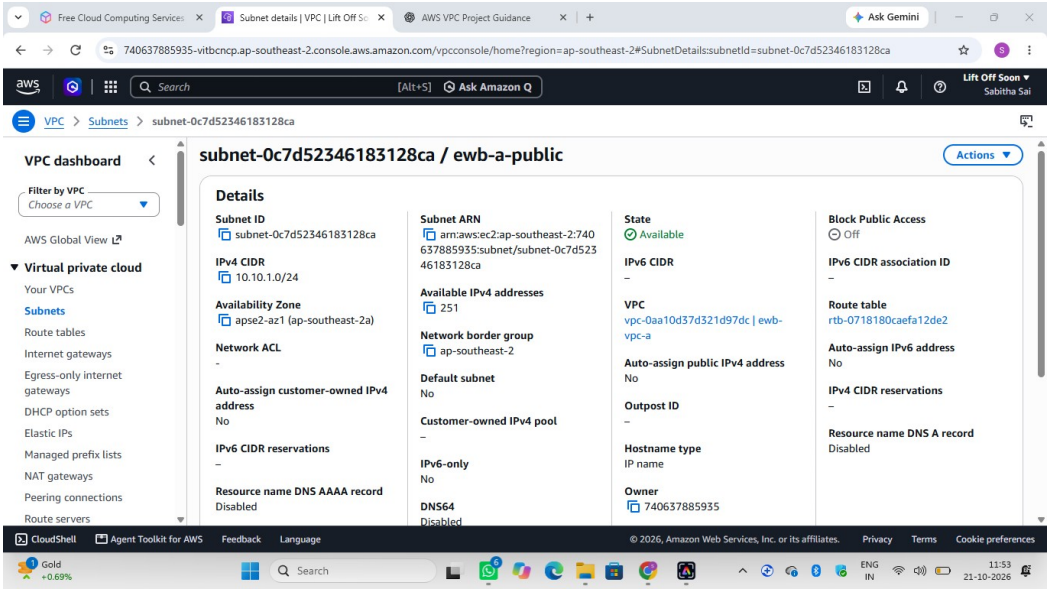


Figure 4.1b: Public subnet ewb-a-public (subnet-0c7d52346183128ca) details

4.2 ewb-rt-a-private (rtb-088c5bfb543bc304f)

Associated subnet: Private subnet A (10.10.2.0/24)

Destination	Target
10.10.0.0/16	local
10.20.0.0/16	Peering (pcx-0e4cab8b7bee709a3)

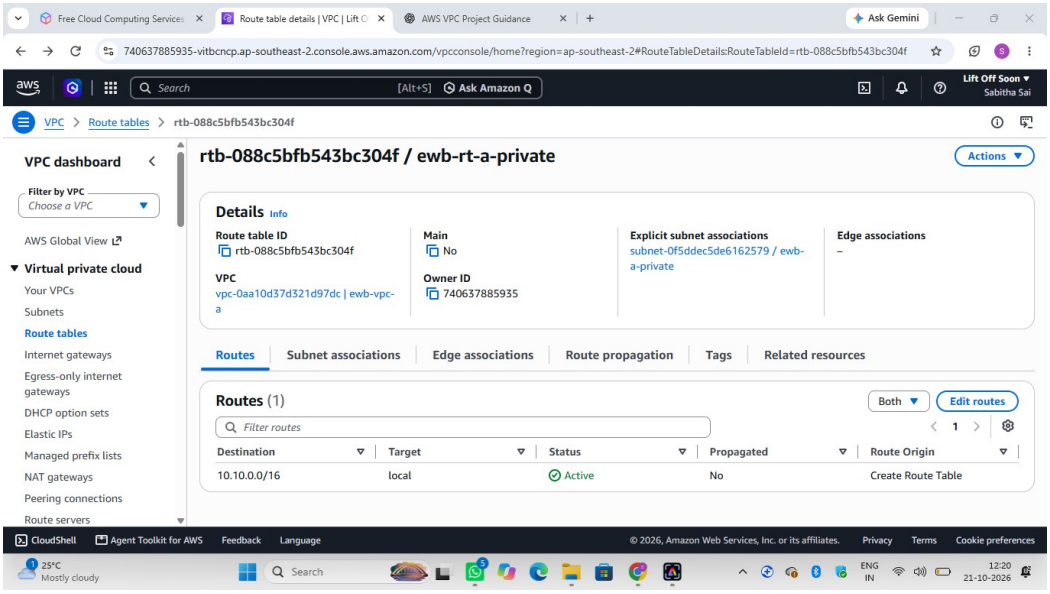


Figure 4.2a: ewb-rt-a-private – Routes tab

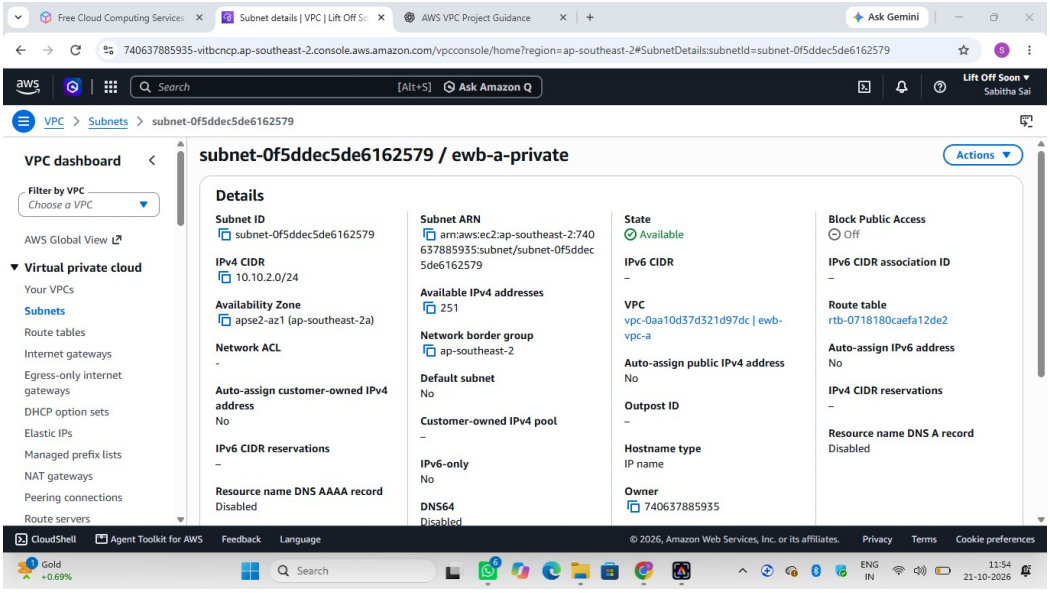


Figure 4.2b: Private subnet ewb-a-private (subnet-0f5ddec5de6162579) details

4.3 ewb-rt-b-private (rtb-07cd76078c961ea32)

Associated subnet: Private subnet B (10.20.1.0/24)

Destination	Target
10.20.0.0/16	local
10.10.0.0/16	Peering (pcx-0e4cab8b7bee709a3)

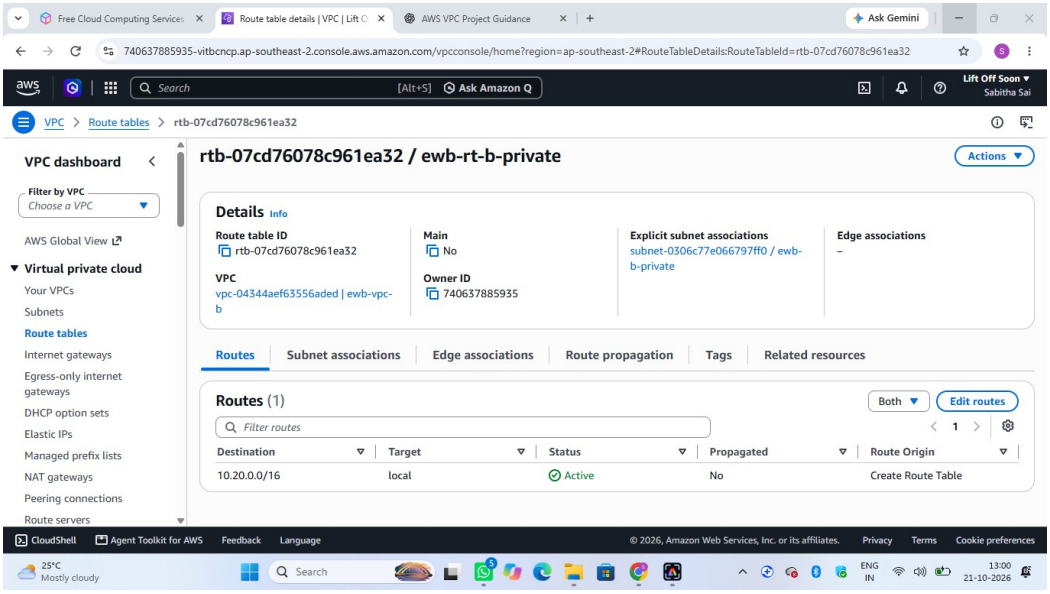


Figure 4.3a: ewb-rt-b-private – Routes tab

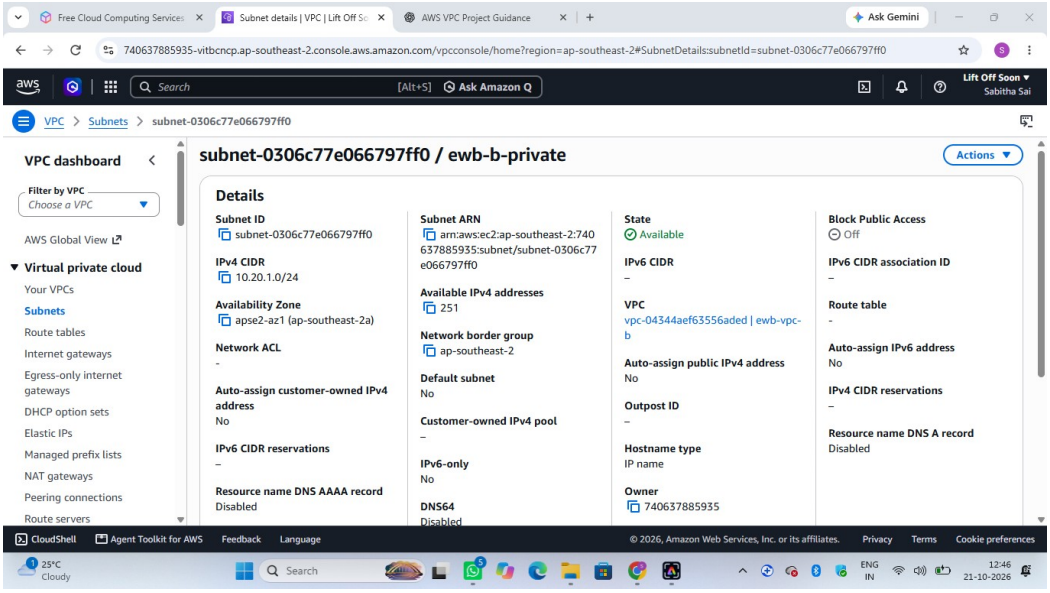


Figure 4.3b: Private subnet ewb-b-private (subnet-0306c77e066797ff0) details

5. Peering and Security Groups

5.1 VPC Peering Connection

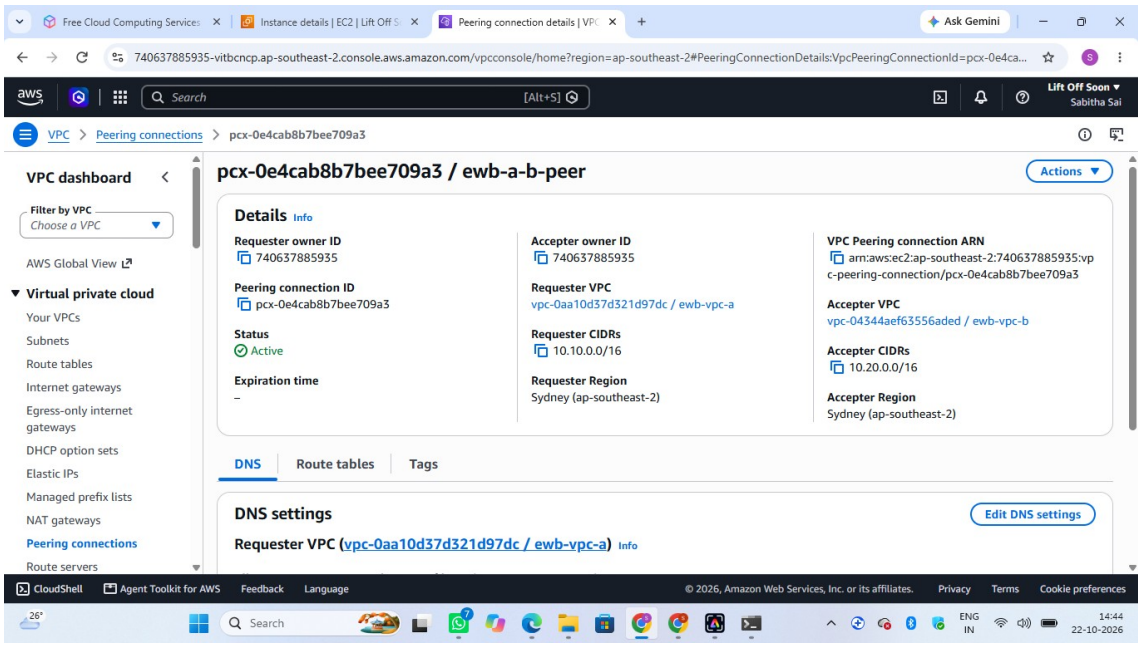


Figure 5.1: VPC peering connection ewb-a-b-peer (pcx-0e4cab8b7bee709a3) – Status: Active

5.2 Security Group Inbound Rules

The tables below list the inbound rules configured on each security group, followed by a console screenshot of the same rules.

Bastion security group (ewb-sg-bastion)

Type	Port	Source	Purpose
SSH	22	157.51.102.135/32	Admin SSH to Bastion only from your own IP

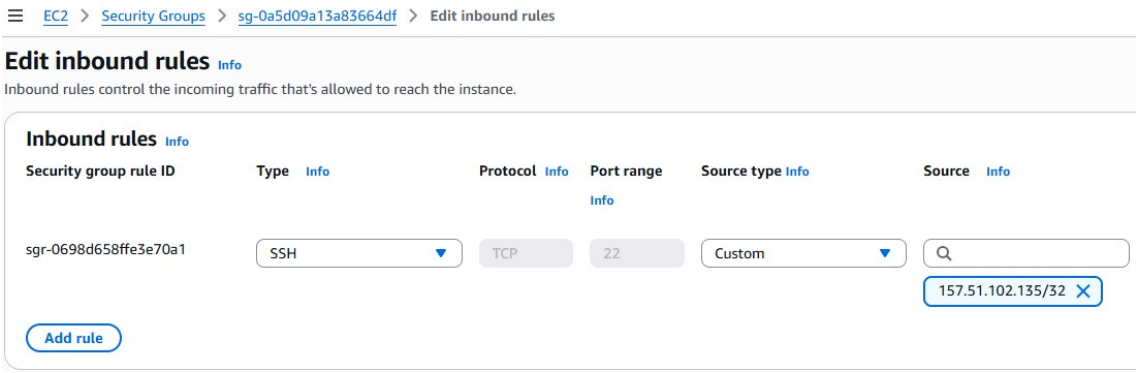


Figure 5.2a: Bastion SG (ewb-sg-bastion) – Inbound rules

App-A security group (ewb-sg-app-a)

Type	Port	Source	Purpose
SSH	22	10.10.1.10/32 (or Bastion SG)	SSH from Bastion
All ICMP – IPv4 / Echo Request	N/A	10.20.1.10/32	Ping from App-B

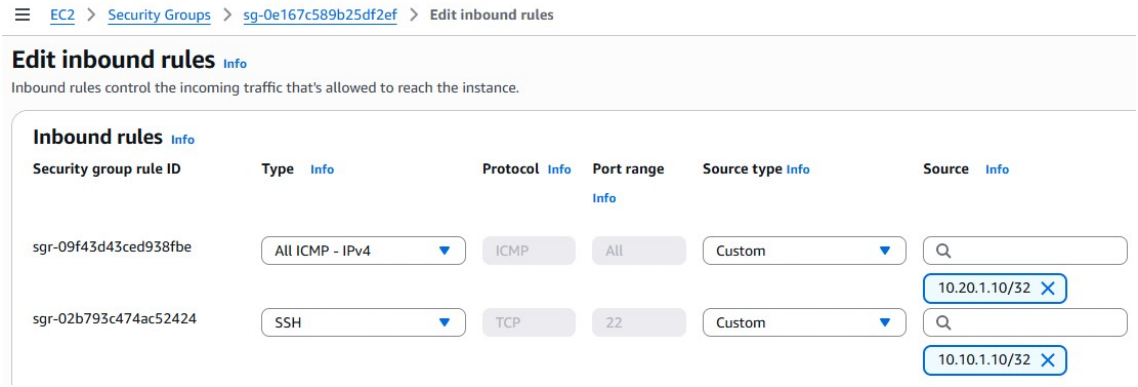


Figure 5.2b: App-A SG (ewb-sg-app-a) – Inbound rules

App-B security group (ewb-sg-app-b)

Type	Port	Source	Purpose
SSH	22	10.10.1.10/32	SSH from Bastion over peering
All ICMP – IPv4 / Echo Request	N/A	10.10.2.10/32	Ping from App-A

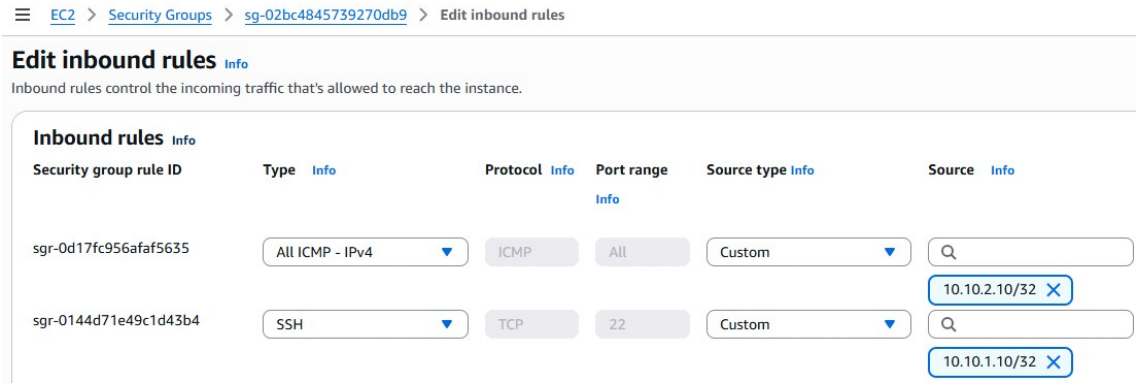


Figure 5.2c: App-B SG (ewb-sg-app-b) – Inbound rules

5.3 No Public IPv4 on Private Instances

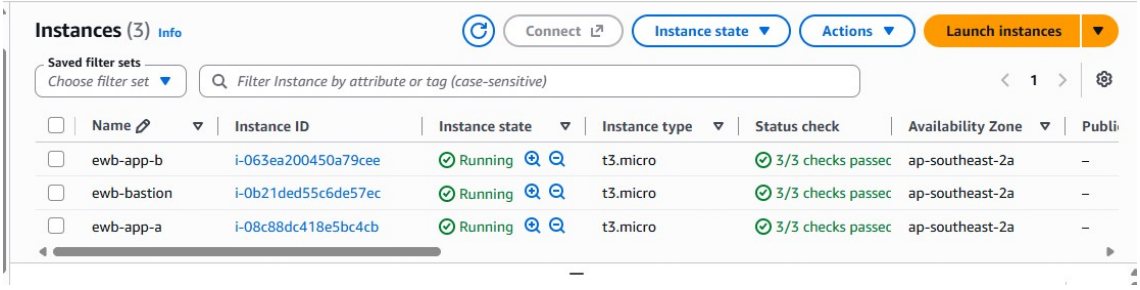


Figure 5.3: EC2 console – ewb-app-a and ewb-app-b with no public IPv4 address

6. Testing and Results

6.1 App-A → App-B ping success

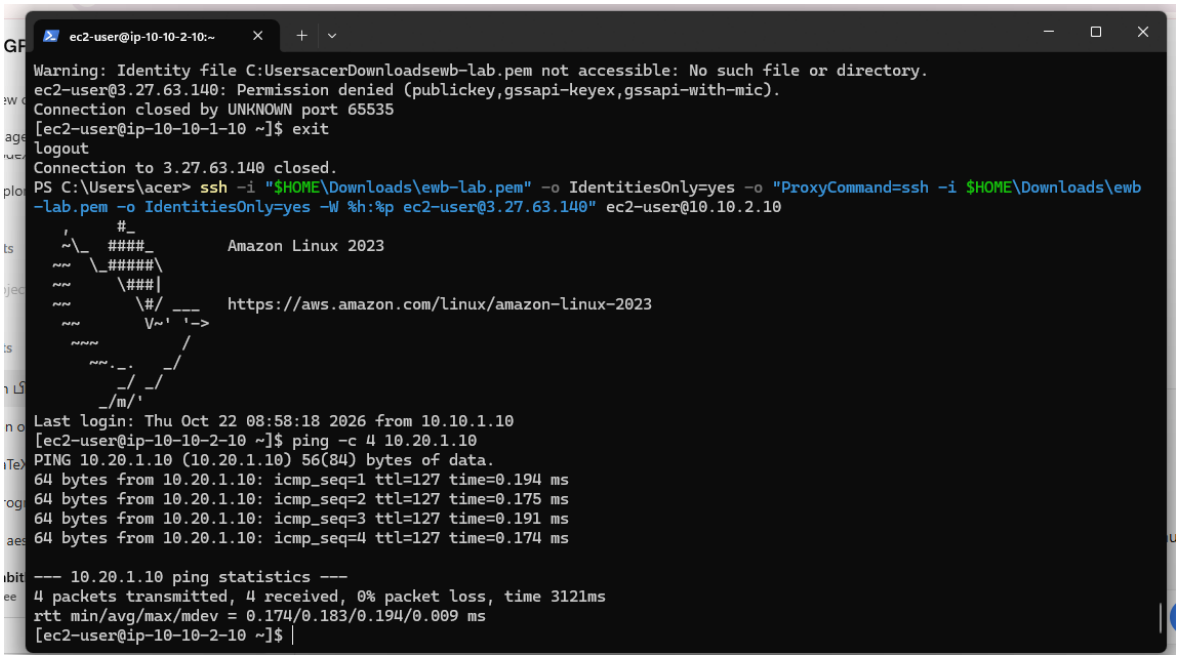


Figure 6.1: App-A (10.10.2.10) → App-B (10.20.1.10) ping success, 0% packet loss

Caption	Details
Test name	App-A → App-B ping success
Host	App-A (10.10.2.10)
Expected result	Ping to 10.20.1.10 succeeds with 0% packet loss.
Actual result	Ping from App-A (10.10.2.10) to 10.20.1.10 succeeded: 4 packets transmitted, 4 received, 0% packet loss (avg RTT 0.183 ms).

6.2 App-B → App-A reverse ping success

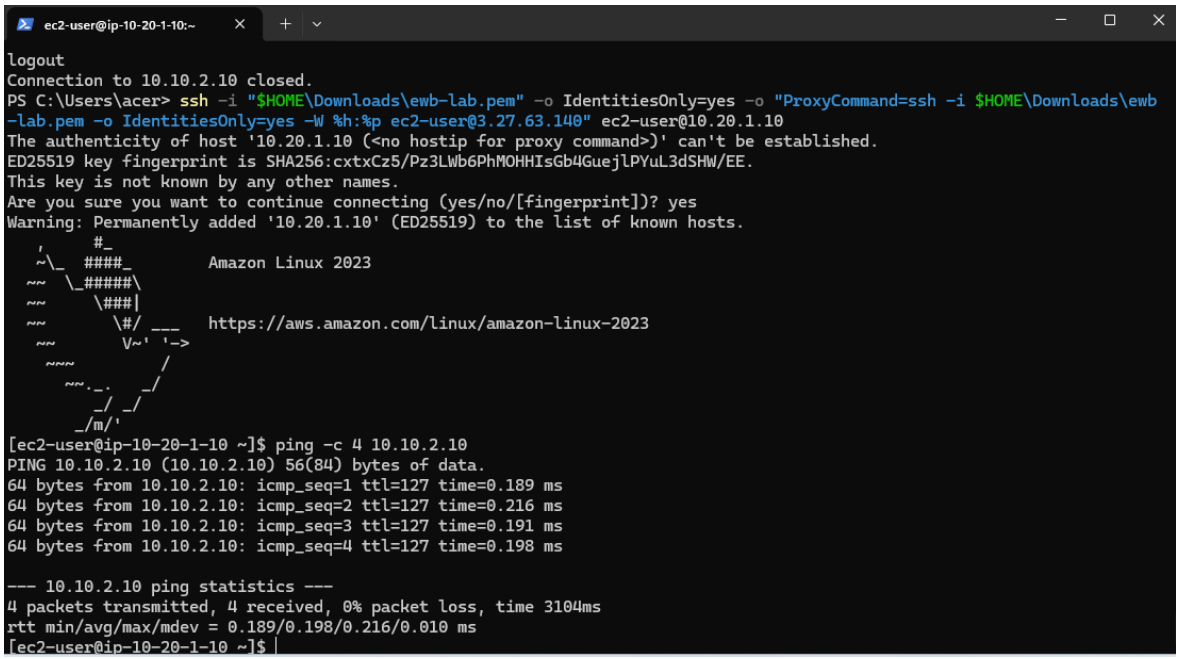


Figure 6.2: App-B (10.20.1.10) → App-A (10.10.2.10) reverse ping success, 0% packet loss

Caption	Details
Test name	App-B → App-A reverse ping success
Host	App-B (10.20.1.10)
Expected result	Ping to 10.10.2.10 succeeds with 0% packet loss.
Actual result	Ping from App-B (10.20.1.10) to 10.10.2.10 succeeded: 4 packets transmitted, 4 received, 0% packet loss (avg RTT 0.198 ms).

6.3 App-B SSH through Bastion

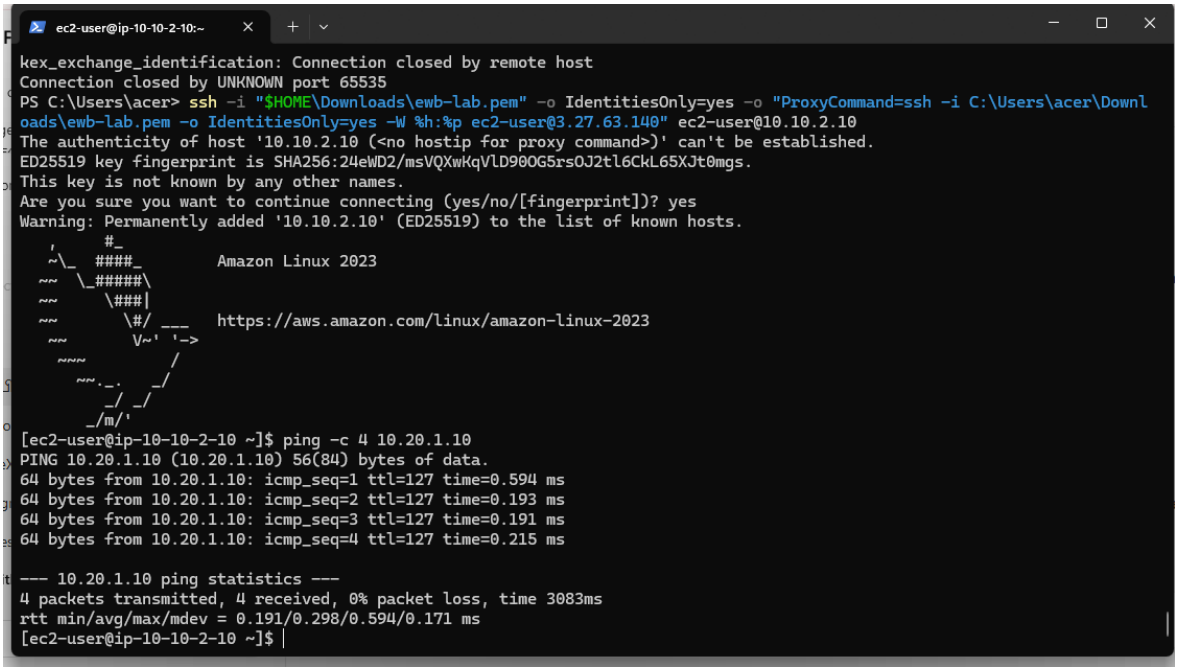


Figure 6.3: SSH to the private host through the Bastion (ProxyCommand via 3.27.63.140), then ping to 10.20.1.10 with 0% packet loss

Caption	Details
Test name	App-B SSH through Bastion
Host	Bastion (10.10.1.10) → App-B (10.20.1.10)
Expected result	SSH session to App-B opens from the Bastion using the private IP.
Actual result	SSH through the Bastion public IP (3.27.63.140) using ProxyCommand opened a session on the private host using its private IP; ping to 10.20.1.10 from that session succeeded with 0% packet loss (avg RTT 0.298 ms).

6.4 Ping failure after ICMP rule removal

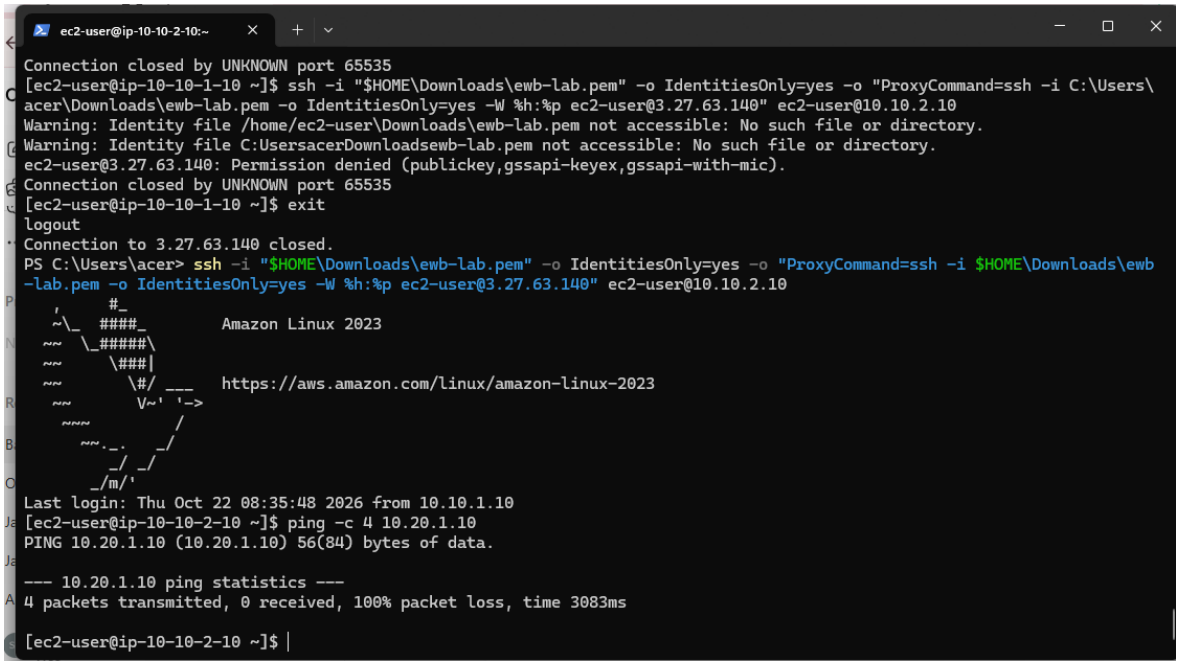


Figure 6.4: App-A (10.10.2.10) → App-B (10.20.1.10) ping failure, 100% packet loss after ICMP rule removal

Caption	Details
Test name	Ping failure after ICMP rule removal
Host	App-A (10.10.2.10) → App-B (10.20.1.10)
Expected result	Ping fails after the ICMP inbound rule is removed from the App-B SG.
Actual result	Ping from App-A (10.10.2.10) to 10.20.1.10 failed: 4 packets transmitted, 0 received, 100% packet loss after the ICMP rule was removed from the App-B SG.

6.5 Ping success after ICMP rule restore

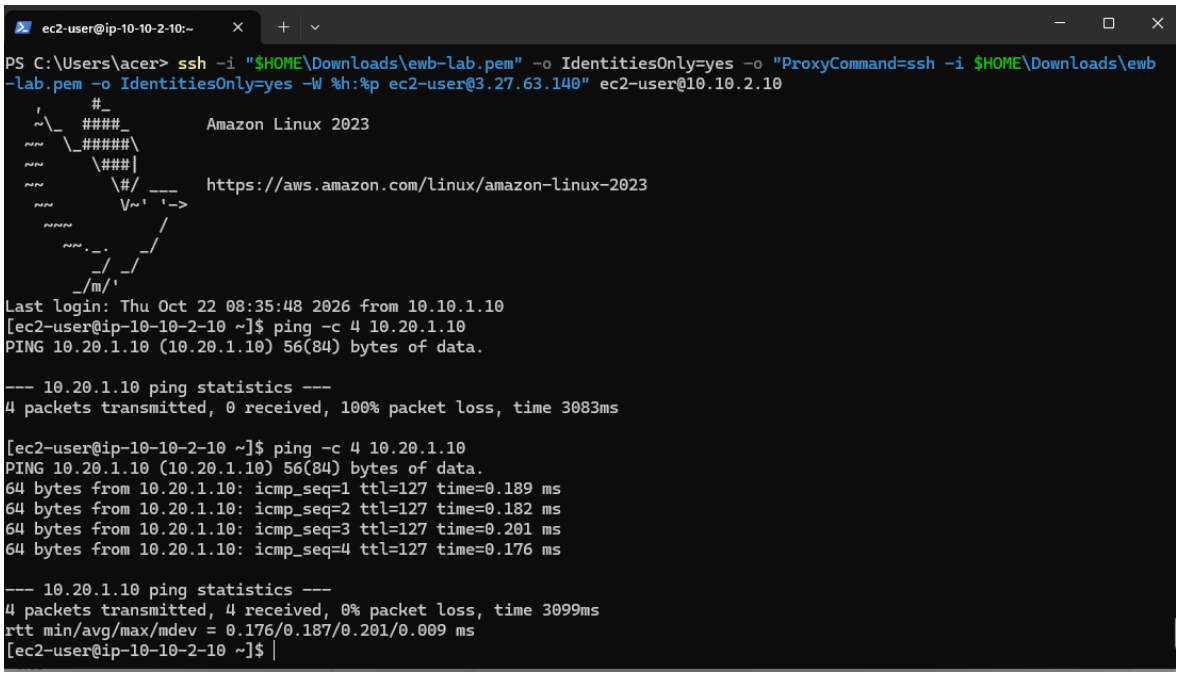


Figure 6.5: App-A (10.10.2.10) → App-B (10.20.1.10) ping success after ICMP rule restore, 0% packet loss

Caption	Details
Test name	Ping success after ICMP rule restore
Host	App-A (10.10.2.10) → App-B (10.20.1.10)
Expected result	Ping succeeds again after the ICMP rule is restored.
Actual result	Ping from App-A (10.10.2.10) to 10.20.1.10 succeeded again: 4 packets transmitted, 4 received, 0% packet loss (avg RTT 0.187 ms).

7. Network Flow Explanation

App-A (10.10.2.10) sends ICMP Echo Requests to App-B (10.20.1.10). The source VPC route table forwards the traffic through the active VPC peering connection. App-B's security group allows the ICMP Echo Request from App-A's private IP. The response returns through VPC B's route table and the peering connection to App-A. The return route and destination security rule are both necessary for successful communication.

SSH access is made through the Bastion public IP. From the Bastion, the private IPs are used to access App-A and App-B.

8. Conclusion and Cleanup Evidence

What was configured

Two VPCs (10.10.0.0/16 and 10.20.0.0/16) with a public subnet and a private subnet in VPC A and a private subnet in VPC B, an Internet Gateway for VPC A, a Bastion host and two private application instances, three route tables, three security groups, and the ewb-a-b-peer VPC peering connection.

How VPC peering worked

After the peering request was accepted and routes to the other VPC's CIDR were added on both sides, instances communicated over private IPs. Peering is not transitive and does not work without routes in both VPCs.

Why return routes and security groups matter

A route is needed in each VPC: one for the request and one for the reply. Without the return route, replies cannot reach the source. The destination security group must also allow the traffic, otherwise it is dropped even if routing is correct.

Learning from the controlled failure test

ICMP was used to verify network connectivity between App-A and App-B across the VPC peering connection. Successful ping responses confirmed that the required routes and ICMP security group rules were configured correctly.

When the inbound ICMP rule was removed from the App-B security group (ewb-sg-app-b), the ping from App-A (10.10.2.10) to App-B (10.20.1.10) immediately failed with 100% packet loss (4 packets transmitted, 0 received). The route tables and the peering connection were unchanged, so the cause was identified as the missing ICMP rule in the security group, which blocked the echo requests.

After the ICMP inbound rule was added back to the App-B security group, the same ping succeeded again with 0% packet loss (4 packets transmitted, 4 received). This showed that VPC peering and routing alone are not enough: the security groups must also allow the required traffic in order for communication to work.

Cleanup evidence

Cleanup order followed: terminate instances → delete peering → delete route table entries/route tables → detach and delete IGW → delete subnets → delete security groups → delete VPCs

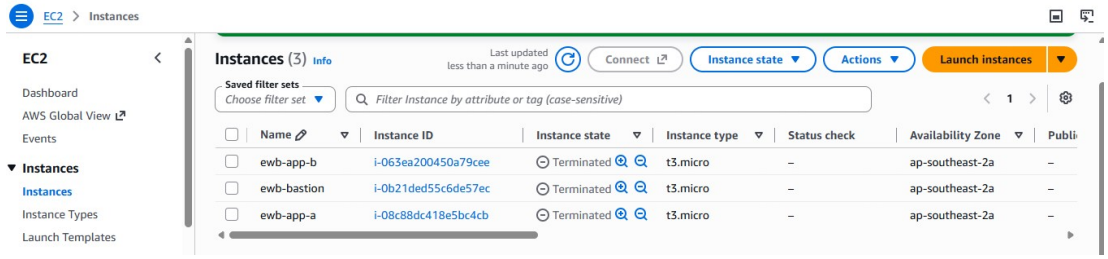


Figure 8.1: EC2 console – ewb-bastion, ewb-app-a and ewb-app-b in Terminated state

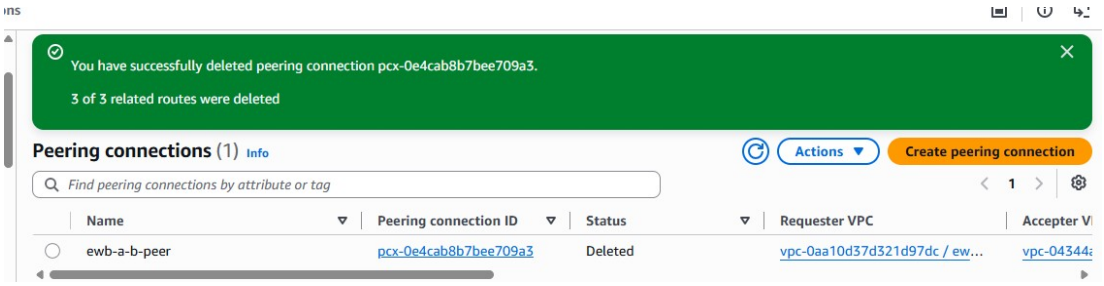


Figure 8.2: VPC peering connection ewb-a-b-peer (pcx-0e4cab8b7bee709a3) – Status: Deleted

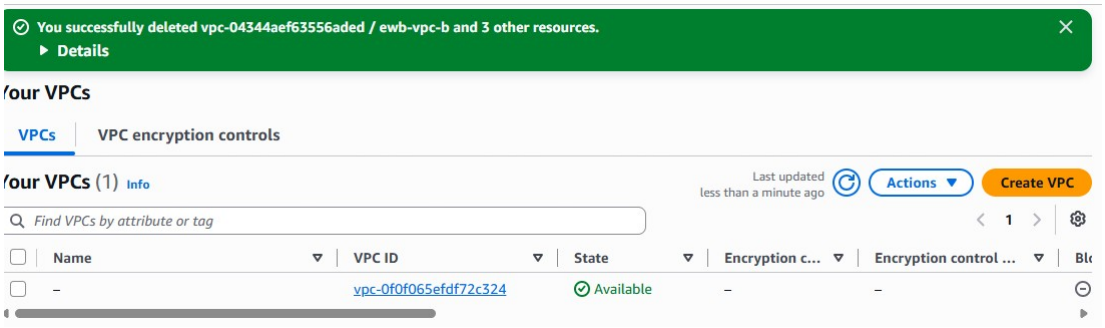


Figure 8.3: VPC console – ewb-vpc-a and ewb-vpc-b deleted, no longer listed